

Step-by-step Wolfram Derivations of the Statistical Pole

Propositions 8.1–8.6 of *The Electromagnetic Sphere*

Fisher–Rao geometry, dual connections, max-entropy selection, and the residual Weyl symmetry

Verification note for the ES monograph (Part I)

June 18, 2026

Abstract

This note works through the entire statistical-pole chapter of the Electromagnetic Sphere monograph as a sequence of reproducible Wolfram Language computations. The candidate clocks form the 2-simplex Δ^2 ; its Fisher–Rao metric has constant Gaussian curvature $K = \frac{1}{4}$ and, under the Born amplitude map $p_i = x_i^2$, becomes the round metric $4 \sum_i dx_i^2$ (8.1). The amplitude map sends the diagonal condensate to the uniform distribution and an axis condensate to a deterministic one (8.2). The simplex is dually flat: the m -connection vanishes in mixture coordinates, the e -connection vanishes in exponential coordinates, the dual-flatness relation holds, and the Legendre map returns $\theta^i = \log(p_i/p_3)$ (8.3). The negative Shannon entropy is the canonical potential and its Bregman divergence is the Kullback–Leibler divergence, nonnegative and zero iff $p = q$ (8.4). The entropy has a unique critical point at the uniform distribution, a maximum with negative-definite Hessian and $S = \log 3$ (8.5). The residual symmetry of the Cartan condensate is the Weyl group $W(D_3) = (\mathbb{Z}_2)^2 \rtimes S_3 \cong S_4$ of order 24, the even sign changes, *not* the hyperoctahedral $W(B_3)$ of order 48 (8.6). Status: **[EXACT]** for 8.1, 8.2, 8.3, 8.5, 8.6; **[DERIVED]** for the entropy/KL corollary 8.4.

1 Setup: the clock-weight simplex

The three temporal generators e_1, e_2, e_3 define a three-parameter family of candidate timelike directions; normalizing the nonnegative weights gives the 2-simplex $\Delta^2 = \{(p_1, p_2, p_3) : p_i \geq 0, \sum_i p_i = 1\}$, each interior point a probability distribution over the three clocks. Throughout we use the chart $p_3 = 1 - p_1 - p_2$. The amplitude (Born) map relates the simplex to the Cartan condensate $\Phi = \sum_i \phi_i \omega^i$ via $p_i = \phi_i^2/|\phi|^2$, $\xi_i = \sqrt{p_i}$. All computations below are carried out symbolically.

2 Proposition 8.1: the statistical sphere

Proposition 1 (The statistical sphere, monograph 8.1). *The Fisher–Rao metric $ds^2 = \sum_i dp_i^2/p_i$ on Δ^2 has constant Gaussian curvature $K = \frac{1}{4}$; under $p_i = x_i^2$ (with $\sum_i x_i^2 = 1$) it becomes $4 \sum_i dx_i^2$, the round sphere metric. **[EXACT]***

Curvature by the Levi–Civita construction

We form the metric in the chart, compute Christoffel symbols, the Riemann tensor, and $K = R_{1212}/\det g$, then check the substitution residual.

Wolfram Language input

```

p3 = 1 - p1 - p2;
gF = {{1/p1 + 1/p3, 1/p3}, {1/p3, 1/p2 + 1/p3}};
coords = {p1, p2}; ginv = Inverse[gF];
Christoffel = Table[(1/2) Sum[ginv[[k,1]] (D[gF[[j,1]],coords[[i]]
+ D[gF[[i,1]],coords[[j]]] - D[gF[[i,j]],coords[[1]]]), {1,2}], {k,2},{i,2},{j,2}];
Christoffel = Simplify[Christoffel, Assumptions->{p1>0,p2>0,p1+p2<1}];
Riem = Table[D[Christoffel[[1,i,k]],coords[[j]] - D[Christoffel[[1,i,j]],coords[[k]]
+ Sum[Christoffel[[1,j,m]] Christoffel[[m,i,k]]
- Christoffel[[1,k,m]] Christoffel[[m,i,j]], {m,2}], {1,2},{i,2},{j,2},{k,2}];
Rlow = Table[Sum[gF[[1,m]] Riem[[m,i,j,k]], {m,2}], {1,2},{i,2},{j,2},{k,2}];
R1212 = Simplify[Rlow[[1,2,1,2]], Assumptions->{p1>0,p2>0,p1+p2<1}];
detg = Simplify[Det[gF], Assumptions->{p1>0,p2>0,p1+p2<1}];
Kgauss= Simplify[R1212/detg, Assumptions->{p1>0,p2>0,p1+p2<1}];
xs={x1,x2,x3}; dxs={dx1,dx2,dx3};
dpi=Table[D[xs[[i]]^2,xs[[i]]] dxs[[i]],{i,3}];
residual=Simplify[Sum[dpi[[i]]^2/xs[[i]]^2,{i,3}] - 4 Sum[dxs[[i]]^2,{i,3}]];
{"det g"->detg, "R_{1212}"->R1212, "K = R_{1212}/det g"->Kgauss,
"subst residual"->residual}

```

Literal Wolfram output

```

det g          = 1/(p1 p2 p3)          (= -1/(p1 p2 (-1+p1+p2)))
R_{1212}       = (1/4)/(p1 p2 p3)
K = R1212/det g = 1/4
subst residual = 0

```

(A)–(C)

(A) Anchors. $\det g = 1/(p_1 p_2 p_3)$; $R_{1212} = (1/4)/(p_1 p_2 p_3)$; $K = 1/4$; substitution residual = 0. All **[EXACT]**.

(B) Symmetries. Both $\det g$ and R_{1212} are fully symmetric in (p_1, p_2, p_3) , so the curvature $K = 1/4$ is constant and S_3 -invariant over the whole simplex – a homogeneous space, the positive octant of a round 2-sphere.

(C) Interpretation & limits. The clock-weight space is intrinsically a constant positive-curvature surface: the “all-plus” statistical pole, in contrast to the Lorentzian all-minus pole. The result is exact and global on the open simplex; the boundary (where some $p_i \rightarrow 0$) is the deterministic locus and is treated as a limit in 8.2.

3 Proposition 8.2: the two kernels are the extremal distributions

Proposition 2 (Extremal distributions, monograph 8.2). *Under the amplitude map the diagonal condensate $\phi \propto (1, 1, 1)$ is the uniform distribution $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, while an axis condensate $\phi \propto (1, 0, 0)$ is deterministic $(1, 0, 0)$. **[EXACT]***

Wolfram Language input

```

ampMap[phi_] := phi^2 / Total[phi^2];
pDiag = ampMap[{1,1,1}];
pAxis = ampMap[{1,0,0}];

```

```

Sdiag = -Total[pDiag Log[pDiag]];
Saxis = Limit[-q Log[q] - 2 ((1-q)/2) Log[(1-q)/2], q -> 1];
{"(1,1,1)->p"->pDiag, "(1,0,0)->p"->pAxis,
 "S(uniform)"->Simplify[Sdiag], "S(uniform)==Log3"->Simplify[Sdiag==Log[3]],
 "S(deterministic) limit"->Saxis}

```

Literal Wolfram output

```

{"(1,1,1)->p" -> {1/3, 1/3, 1/3}, "(1,0,0)->p" -> {1, 0, 0},
 "S(uniform)" -> Log[3], "S(uniform)==Log3" -> True,
 "S(deterministic) limit" -> 0}

```

(A)–(C)

(A) Anchors. $(1, 1, 1) \mapsto (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ with $S = \log 3$; $(1, 0, 0) \mapsto (1, 0, 0)$ with $S \rightarrow 0$. **[EXACT]**

(B) Symmetries. The map $p_i = \phi_i^2/|\phi|^2$ is invariant under $\phi_i \mapsto -\phi_i$ (the Born square) and equivariant under S_3 permutations. The diagonal is the unique S_3 -fixed point; the axis points are the orbit of maximal isotropy breaking.

(C) Interpretation & limits. The democratic kernel $\hat{W}$ is the maximum-entropy image and an axis clock the minimum-entropy (deterministic) image; this is the kinematic content that 8.5 turns into a selection principle. The deterministic value $S = 0$ is reached only in the boundary limit $q \rightarrow 1$, computed here as a limit rather than a value on the open simplex.

4 Proposition 8.3: the dually flat structure

Proposition 3 (Dually flat structure, monograph 8.3). Δ^2 is dually flat: in mixture coordinates $\eta_i = p_i$ the m -connection $\nabla^{(-1)}$ is flat; in exponential coordinates $\theta^i = \log(p_i/p_3)$ the e -connection $\nabla^{(+1)}$ is flat; the dual-flatness relation $\partial_k g_{ij}^F = \Gamma_{ki,j}^{(+1)} + \Gamma_{kj,i}^{(-1)}$ holds, and $\theta^i = \partial\psi^*/\partial\eta_i$ inverts the Legendre pair. **[EXACT]**

Wolfram Language input

```

alphaConn[l1_List, pr_List, xi_List, a_] := Table[
  Sum[pr[[s]] (D[l1[[s]],xi[[i]],xi[[j]]) + (1-a)/2 D[l1[[s]],xi[[i]] D[l1[[s]],xi[[j]]])
    D[l1[[s]],xi[[k]]], {s,Length[pr]}], {i,Length[xi]}, {j,Length[xi]}, {k,Length[xi]}];
fisher[l1_List, pr_List, xi_List] := Table[
  Sum[pr[[s]] D[l1[[s]],xi[[i]] D[l1[[s]],xi[[j]]], {s,Length[pr]}], {i,Length[xi]}, {j,Length[
    xi]}];
(* (a) eta = p coordinates *)
pE={p1,p2,1-p1-p2}; l1E=Log/@pE; xiE={p1,p2};
GmM1eta=Simplify[alphaConn[l1E,pE,xiE,-1], Assumptions->{0<p1<1,0<p2<1,p1+p2<1}];
gEta=Simplify[fisher[l1E,pE,xiE], Assumptions->{0<p1<1,0<p2<1,p1+p2<1}];
GpP1eta=Simplify[alphaConn[l1E,pE,xiE,+1], Assumptions->{0<p1<1,0<p2<1,p1+p2<1}];
(* (b) theta = log(p_i/p3) coordinates *)
Z=1+Exp[t1]+Exp[t2]; pT={Exp[t1]/Z,Exp[t2]/Z,1/Z}; l1T=Log/@pT; xiT={t1,t2};
GpP1theta=Simplify[alphaConn[l1T,pT,xiT,+1]];
(* dual flatness in eta coords *)
lhsDF=Table[D[gEta[[i,j]],xiE[[k]]],{k,2},{i,2},{j,2}];
rhsDF=Table[GpP1eta[[k,i,j]]+GmM1eta[[k,j,i]],{k,2},{i,2},{j,2}];
DF=Simplify[lhsDF-rhsDF, Assumptions->{0<p1<1,0<p2<1,p1+p2<1}];
(* Legendre *)

```

```
psiStar=p1 Log[p1]+p2 Log[p2]+(1-p1-p2) Log[1-p1-p2];
Leg=Simplify[{D[psiStar,p1],D[psiStar,p2]} - {Log[p1/(1-p1-p2)],Log[p2/(1-p1-p2)]},
  Assumptions->{0<p1<1,0<p2<1,p1+p2<1}];
{"m-conn Gamma^(-1) in eta (all 0?)"->Union@Flatten[GmM1eta],
 "e-conn Gamma^(+1) in theta (all 0?)"->Union@Flatten[GpP1theta],
 "dual-flatness residual (all 0?)"->Union@Flatten[DF],
 "Legendre dpside/deta - log(p_i/p3)"->Leg}
```

Literal Wolfram output

```
m-conn Gamma^(-1) in eta (all 0?) = {0}
e-conn Gamma^(+1) in theta (all 0?) = {0}
dual-flatness residual (all 0?) = {0}
Legendre dpside/deta - log(p_i/p3) = {0, 0}
Fisher metric in eta coords = {{(1-p2)/(p1 p3), 1/p3}, {1/p3, (1-p1)/(p2 p3)}}
```

(A)–(C)

(A) Anchors. $\Gamma^{(-1)}|_\eta = \{0\}$; $\Gamma^{(+1)}|_\theta = \{0\}$; dual-flatness residual = $\{0\}$; $\partial\psi^*/\partial\eta_i - \log(p_i/p_3) = \{0, 0\}$. All **[EXACT]**.

(B) Symmetries. The two flat connections are exchanged by the Legendre duality $\eta \leftrightarrow \theta$; their arithmetic mean is the Levi–Civita connection of the Fisher metric. The exponential and mixture coordinate systems are dual affine charts of the same S_3 -symmetric simplex.

(C) Interpretation & limits. *Three independent zero outputs* (m -connection, e -connection, dual-flatness residual) jointly establish dual flatness; one cannot read curvature into a sector whose connection coefficients vanish identically. The Legendre map returns exactly $\theta^i = \log(p_i/p_3)$, identifying the potentials $\psi(\theta) = \log Z = -\log p_3$ and $\psi^*(\eta) = \sum_i p_i \log p_i$ used in 8.4.

5 Corollary 8.4: entropy is the canonical potential; KL is the canonical divergence

Corollary 1 (Entropy/KL, monograph 8.4). *The Legendre potential of the mixture-flat structure is the negative Shannon entropy $\psi^*(\eta) = \sum_i p_i \log p_i = -S(p)$, and the Bregman divergence it induces is the Kullback–Leibler divergence $D_{\text{KL}}(p \parallel q) = \sum_i p_i \log(p_i/q_i) \geq 0$, zero iff $p = q$. **[DERIVED]***

Wolfram Language input

```
psi[a_,b_] := a Log[a] + b Log[b] + (1-a-b) Log[1-a-b];
grad = {D[psi[u,v],u], D[psi[u,v],v]} /. {u->q1, v->q2};
Bregman = (psi[p1,p2] - psi[q1,q2]) - (grad . ({p1,p2}-{q1,q2}));
KL = p1 Log[p1/q1] + p2 Log[p2/q2] + (1-p1-p2) Log[(1-p1-p2)/(1-q1-q2)];
asm = {0<p1<1,0<p2<1,p1+p2<1,0<q1<1,0<q2<1,q1+q2<1};
diffSym = FullSimplify[Bregman - KL, Assumptions->asm];
diffNum = Table[{p1n,p2n,q1n,q2n}=RandomReal[{.05,.4},4];
  N[(Bregman-KL)/.{p1->p1n,p2->p2n,q1->q1n,q2->q2n}], {6}];
KLval = (KL /. {p1->.2,p2->.3,q1->.4,q2->.25}) // N;
KLeq = (KL /. {p1->.3,p2->.3,q1->.3,q2->.3}) // N;
{"Bregman - KL (FullSimplify)"->diffSym, "Bregman - KL (6 random)"->diffNum,
 "KL(p!=q)"->KLval, "KL(p=q)"->KLeq}
```

Literal Wolfram output

```
Bregman - KL (FullSimplify) = 0
Bregman - KL (6 random)    = {0., 1.11022*10^-16, 0., 0., -2.22045*10^-16, 0.}
KL(p!=q)                   = 0.0944045
KL(p=q)                    = -4.44089*10^-17    (= 0 to machine precision)
```

(A)–(C)

(A) Anchors. Bregman–KL = 0 symbolically and $\sim 10^{-16}$ at six random points; $D_{\text{KL}}(p \neq q) = 0.094\dots > 0$, $D_{\text{KL}}(p = q) = 0$. The identity is **[EXACT]**; the corollary’s *reading* (entropy as canonical potential) is **[DERIVED]**.

(B) Symmetries. The Bregman construction is invariant under affine reparametrization of the dual coordinates; the resulting D_{KL} inherits the information-geometric invariance and the Gibbs nonnegativity $D_{\text{KL}} \geq 0$ with equality exactly on the diagonal $p = q$.

(C) Interpretation & limits. The Shannon entropy is *supplied* by the pole’s own dually flat geometry rather than imposed: the negative entropy is the canonical potential, the D_{KL} divergence its Bregman divergence. The corollary is marked **[DERIVED]** because it interprets the exact zero of Bregman–KL as the entropy identity; the numeric machine-precision residuals confirm the symbolic zero is not an artifact.

6 Proposition 8.5: the kernel is the entropy maximum

Proposition 4 (Entropy maximum, monograph 8.5). *The Shannon entropy $S(p) = -\sum_i p_i \log p_i$ on Δ^2 has a unique critical point, the uniform distribution $p_1 = p_2 = p_3 = \frac{1}{3}$, which is a maximum (negative-definite Hessian), with $S = \log 3$; this is the kernel $\hat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$. **[EXACT]***

Wolfram Language input

```
Sc = -(p1 Log[p1] + p2 Log[p2] + (1-p1-p2) Log[1-p1-p2]);
stat = Simplify[{D[Sc,p1], D[Sc,p2]}];
sol = Solve[{D[Sc,p1]==0, D[Sc,p2]==0}, {p1,p2}, Reals];
Hess = {{D[Sc,{p1,2}], D[Sc,p1,p2]}, {D[Sc,p2,p1], D[Sc,{p2,2}]}} /. {p1->1/3,p2->1/3};
HessEigs = Eigenvalues[Hess];
Smax = Sc /. {p1->1/3, p2->1/3};
{"stationarity eqs (=0)"->stat, "unique solution"->sol,
 "Hessian at uniform"->Hess, "Hessian eigenvalues"->HessEigs, "S_max"->Smax}
```

Literal Wolfram output

```
stationarity eqs (=0) = {-Log[p1] + Log[1 - p1 - p2], Log[1 - p1 - p2] - Log[p2]}
unique solution      = {{p1 -> 1/3, p2 -> 1/3}}
Hessian at uniform   = {{-6, -3}, {-3, -6}}
Hessian eigenvalues  = {-9, -3}      (negative-definite)
S_max                = Log[3]
```

(A)–(C)

(A) Anchors. stationarity $\Rightarrow$ unique $(p_1, p_2) = (\frac{1}{3}, \frac{1}{3})$; Hessian $\begin{pmatrix} -6 & -3 \\ -3 & -6 \end{pmatrix}$ with eigenvalues $\{-9, -3\}$ (both < 0); $S_{\text{max}} = \log 3$. All **[EXACT]**.

(B) **Symmetries.** The stationarity locus and the Hessian are S_3 -symmetric; the unique critical point is the S_3 -fixed centroid, and the negative-definite Hessian (eigenvalues $-9, -3$) certifies it as the global maximum on the compact simplex.

(C) **Interpretation & limits.** The kernel is the democratic diagonal because it is the maximum-entropy point of the clock simplex – the least committal, most symmetric clock. This is the *kinematic* selection principle; the *dynamical* mechanism that drives the condensate there depends on the couplings (g, v, g_3) , whose magnitudes are **[OPEN]**.

7 Proposition 8.6: the residual symmetry is $W(D_3)$, not B_3

Proposition 5 (Residual symmetry, monograph 8.6). *The residual symmetry of the Cartan condensate $\Phi = \sum_i \phi_i \omega^i$ is the Weyl group $W(D_3) = (\mathbb{Z}_2)^2 \rtimes S_3 \cong S_4$ of order 24 (the even sign changes), not the hyperoctahedral $W(B_3) = (\mathbb{Z}_2)^3 \rtimes S_3$ of order 48. **[EXACT]***

Wolfram Language input

```
perms = Permutations[{1,2,3}];
permMat[pi_] := Table[If[pi[[j]]==i,1,0],{i,1,3},{j,3}];
signMats = DiagonalMatrix /@ Tuples[{1,-1},3];
B3 = DeleteDuplicates[Flatten[Table[s.permMat[pi], {s,signMats},{pi,perms}],1]];
D3 = Select[B3, (Times @@ (Total /@ #)) == 1 &]; (* even sign changes *)
evenSigns = Select[Tuples[{1,-1},3], (Times@@#)==1 &];
closedQ[grp_] := SubsetQ[grp, DeleteDuplicates[Flatten[Outer[Dot,grp,grp,1],1]]];
(* element-order multisets: a stronger isomorphism test than |G| alone *)
matOrder[m_] := Module[{k=1,p=m,Idd=IdentityMatrix[3]}, While[p!=Idd && k<50, p=p.m; k++]; k];
D3orders = Sort[Tally[matOrder /@ D3]];
S4orders = Sort[Tally[PermutationOrder /@ GroupElements[SymmetricGroup[4]]]];
{"|W(B3)|"->Length[B3], "|W(D3)|"->Length[D3],
 "even-sign subgroup order"->Length[evenSigns], "|(Z2)^2 |x| S3|"->Length[evenSigns]*Length[
 perms],
 "|S4|"->GroupOrder[SymmetricGroup[4]], "B3 closed"->closedQ[B3], "D3 closed"->closedQ[D3],
 "ratio |B3|/|D3|"->Length[B3]/Length[D3], "W(D3)==W(B3)?"->(Length[D3]==Length[B3]),
 "D3 element-orders"->D3orders, "S4 element-orders"->S4orders,
 "W(D3) ~ S4 (orders match)"->(D3orders===S4orders)}
```

Literal Wolfram output

```
|W(B3)|           = 48
|W(D3)|           = 24
even-sign subgroup order = 4   (= (Z2)^2)
|(Z2)^2 |x| S3|   = 24
|S4|              = 24
B3 closed         = True ;   D3 closed = True
ratio |B3|/|D3|   = 2
W(D3)==W(B3)?    = False
D3 element-orders = {{1,1},{2,9},{3,8},{4,6}}
S4 element-orders = {{1,1},{2,9},{3,8},{4,6}}
W(D3) ~ S4 (orders match) = True
```

(A)–(C)

(A) **Anchors.** $|W(B_3)| = 48$; $|W(D_3)| = 24$; even-sign subgroup order $4 = (\mathbb{Z}_2)^2$; $|(\mathbb{Z}_2)^2 \rtimes S_3| = 24 = |S_4|$; both groups closed; ratio 2; $W(D_3) \neq W(B_3)$; matching element-order multisets

$\{1:1, 2:9, 3:8, 4:6\}$ confirm $W(D_3) \cong S_4$. All **[EXACT]**.

(B) Symmetries. $W(D_3)$ is the kernel of the product-of-signs homomorphism $(\mathbb{Z}_2)^3 \rightarrow \mathbb{Z}_2$ extended by S_3 : the even sign changes. The matching element-order multisets are a stronger isomorphism certificate than the order alone, fixing $W(D_3) \cong S_4$.

(C) Interpretation & limits. The hyperoctahedral B_3 would include odd sign changes – a single reflection $\phi_i \mapsto -\phi_i$ – i.e. a parity the chiral structure forbids. Only the even sign changes (D_3) preserve the condensate’s chiral content. Assuming B_3 would smuggle in a parity the theory does not possess. The group-theory facts are exact and complete.

8 Summary anchor table

Result	Statement	Decisive literal anchor	Status
8.1	Fisher curvature	$K = R_{1212}/\det g = 1/4$; residual = 0	[EXACT]
8.2	extremal distributions	$(1, 1, 1) \rightarrow (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$; $(1, 0, 0) \rightarrow (1, 0, 0)$	[EXACT]
8.3	dual flatness	$\Gamma^{(-1)} _{\eta=0}, \Gamma^{(+1)} _{\theta=0}, \text{DF}=0, \text{Legendre}=0$	[EXACT]
8.4	entropy / KL	Bregman–KL= 0; $D_{\text{KL}} \geq 0$, = 0 iff $p=q$	[DERIVED]
8.5	max-entropy kernel	unique $(\frac{1}{3}, \frac{1}{3})$; Hess eigs $\{-9, -3\}$; $S=\log 3$	[EXACT]
8.6	residual Weyl group	$ D_3 =24 \neq 48= B_3 $; $D_3 \cong S_4$	[EXACT]

Summary. The candidate clocks form the 2-simplex Δ^2 , whose Fisher–Rao geometry is a constant-curvature sphere ($K = \frac{1}{4}$, residual 0). The amplitude map sends the diagonal condensate to the uniform distribution ($S = \log 3$) and an axis to a deterministic one ($S = 0$). The simplex is dually flat (three vanishing connection/residual checks, Legendre map exact), the negative entropy is the canonical potential and KL the canonical divergence (Bregman–KL= 0, $D_{\text{KL}} \geq 0$). The entropy has a unique negative-definite maximum at the uniform distribution ($S = \log 3$), the kernel $\hat{W}$. The residual symmetry is $W(D_3) \cong S_4$ of order 24 (even sign changes), *not* the parity-containing $W(B_3)$ of order 48. All verified symbolically; the dynamical realization of the selection depends on the open couplings.